Common Cause, Not Channel: The Determinability Boundary as an Observational Structure in Event Density

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Abstract

The determinability-boundary measurement found a positive within-stratum mutual information (`Delta > 0`) that is **observable-dependent** — so the scalar `Delta` is not an intrinsic quantity. Channel *capacity* — the supremum of transmitted information over all inputs and readouts — is observable-independent by construction, so it is the natural candidate to rescue a positive intrinsic invariant. We test it with a coding experiment: encode a `K`-ary message into region A's left-half initial condition, evolve the certified substrate, and try to decode it from (i) A's right half (within a stratum) and (ii) region B (across the decoupling boundary). The result is exact and comes from a direct propagation diagnostic: **changing the encoded message leaves the distant region's final state byte-identical — `L2 = 0.000000` — across every baseline (5/5), within a stratum and across the boundary alike**, while the local-response control passes (a region responds to its *own* initial condition, `L2 = 1.8`). The controlled (interventional) channel capacity between distinct regions is therefore **exactly zero everywhere**. Two consequences. **(1)** Capacity yields no positive intrinsic invariant — the only observable-invariant content of the boundary is a zero, confirming and strengthening the prior result. **(2)** The determinability boundary is an **observational (common-cause) structure, not a transmission channel**: the within-stratum correlation `Delta > 0` is shared-origin co-variation (both halves share a run's seed and the front's passage), *not* information flow; the boundary's "severance" is severance of shared origin (independent seeds → independent strata), not of a channel — because there is no channel anywhere. As a corollary, ED's certified **locality** ("no unmediated nonlocal influence") is confirmed empirically as exact channel-zero: a committed fact is strictly local. The boundary's positive content lives entirely at the observational/common-cause level; its interventional content is a universal zero.

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1. Introduction

The determinability-boundary measurement quantified, in bits, how much one region of the certified ED substrate determines about another, and found the within-stratum mutual information `Delta` to be **observable-dependent** — different observables give different `Delta`, so no single `Delta` is intrinsic. Channel capacity is the obvious next move, and it is the *right* candidate for an intrinsic invariant: as the supremum over **all** encodings and readouts it is **observable-independent** by construction, so a positive within-stratum capacity (with a smaller across-boundary value) would be a genuinely intrinsic determinability quantity in a way no single-observable `Delta` can be. Its *failure* is therefore meaningful — it closes the last route to a positive intrinsic scalar. This short paper asks whether capacity delivers, and finds something cleaner and more structural instead. It is a companion to the determinability-boundary measurement, sharpening *what the boundary is.*

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2. Primitive Inputs and Method

Substrate: the certified `Bits/` Σ-rule simulator (20/20 correctness gates), used as-is. The load-bearing property is ED's **locality**: a node's committed state is a function of its local neighborhood only.

The coding experiment. Encode a `K`-ary message into region A's left-half initial condition; evolve the certified substrate; attempt to decode the message from (i) A's right half (a within-stratum channel) and (ii) region B (across the decoupling boundary). The capacity is the best achievable decoding over inputs and readouts.

The decisive diagnostic. Rather than rely on a capacity estimate, we run a direct **propagation diagnostic**: change the encoded message and measure the exact `L2` difference in the distant region's final state. If the message does not propagate, there is nothing to estimate.

Controls. A **local-response control** changes a region's *own* initial condition and confirms it responds (the simulator works); an **across-channel-as-null** control guards the estimator.

No primitive forcing is invoked.

Reproducibility. The certified `Bits/` substrate is run on a grid partitioned into regions **A-left**, **A-right** (the two halves of one stratum) and **B** (across the decoupling boundary). A `K`-ary message is injected by setting region A-left's initial condition to one of `K` distinct values (the diagnostic uses the two extremes, `0.0` and `0.45`); the substrate is evolved to completion and the **final-state arrays** of A-right and B are compared. The propagation metric is the `L2` norm of the difference between the two final-state arrays (`Sigma(x^{msg=0}_{final} - x^{msg=0.45}_{final})^2`), evaluated over **5 baseline configurations** (5/5). The local-response control applies the same `0.45` change to a region's *own* initial condition. Seeds are fixed per run. Scripts: `capacity_experiment.py` (the diagnostic), `capacity.py` (the bias-corrected k-NN MI estimator, used only for the en-route check).

Regime of validity. This result is for the **certified Σ-substrate** and the **specific region geometries and 5 baselines tested**, with specific initial-condition encodings: an exact zero on those cases, with a passing local-response control. It is **not a proof for all possible geometries or ED instantiations** — though ED's locality (§4.3) makes a nonzero result on any *local* partition structurally surprising.

2.5 Load-Bearing Step Audit

Step	Status	Source / justification
13 primitives postulated; certified substrate	P / I	Paper_087; `Bits/` (20/20 gates)
Distant-region propagation is exactly zero (`L2 = 0`, 5/5)	**measured (exact)**	§3 — direct byte-level diagnostic
Local-response control passes (`L2 = 1.8`)	**measured**	§3 — simulator responds to local input
Interventional channel capacity = 0 (within and across)	**D**	§3 — no propagation ⇒ no channel
Estimator bias caught and corrected	**measured / honesty note**	§3 — k-NN decoder MI positively biased; control caught it
Capacity yields no positive intrinsic invariant	**D**	§4.1 — zero everywhere
Boundary is observational (common-cause), not transmission	**interpretation**	§4.2 — within-stratum `Delta` is shared-origin co-variation
ED locality confirmed (exact channel-zero)	**D**	§4.3
A positive intrinsic determinability scalar	**NOT FOUND**	§4.1

All steps P, I, measured, D, interpretation, or explicitly not-found. *The headline is an exact measured zero, not an estimate; the one estimator hazard is reported and was caught by a control.*

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3. The Result: Exactly Zero

The headline did not come from the estimator; it came from the direct propagation diagnostic, and it is exact:

> **Changing the encoded A-left message (0.0 → 0.45) leaves A-right's final state byte-identical — `L2 = 0.000000` — across every baseline tested (5/5).** The message does not propagate to A-right *at all*. Not weakly. Exactly zero. The same holds for region B across the boundary.

"Exact zero" here means bit-for-bit identity of the final-state arrays — `L2 = 0.000000` with no floating-point tolerance and no thresholding; the two runs produce numerically *identical* output, not merely close output.

The **control passes**: A-right *does* respond to its *own* local initial condition (`L2 = 1.8` for the same 0.45 change). So the simulator is working — a region responds to local changes and is *exactly insensitive* to a distant region's input. The logic is then explicit: **if no perturbation to A-left produces any change in A-right or B, the mutual information between input and output is identically zero, so the channel capacity — the supremum of that mutual information over all encodings and readouts — is zero.** The controlled (interventional) channel capacity between distinct regions is therefore **exactly zero — within a stratum and across the boundary alike.** A committed node depends only on its local neighborhood; it carries no information about a distant region's initial condition. There is no controllable channel to have a capacity.

Estimator honesty note. The first decoder-MI pass was *wrong*: a naive leave-one-out k-NN decoder MI is positively biased by `~log_2K - log_2k` when `K >> k`, and it read `~2.8 bits` even across the *decoupled* boundary. The across-channel-as-null control caught it immediately, and a shuffle-null correction fixed it — but once the diagnostic showed propagation is *exactly* zero, no capacity estimate was needed. **To be unambiguous: the MI estimator was *not* used for the final result; the exact-zero propagation diagnostic supersedes it entirely — the conclusion would be identical with no estimator at all.** This is recorded because the trap is easy and the control is what caught it.

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4. What It Means

4.1 Capacity gives no positive intrinsic invariant

At the one genuinely observable-independent level (capacity), the intrinsic quantity is **zero**, and it is zero *everywhere* — within and across. So capacity provides no "within > across" contrast and no canonical positive determinability scalar. The determinability-boundary measurement's conclusion stands and is strengthened: **the only observable-invariant content of the boundary is a zero**, not a scalar.

4.2 The boundary is observational (common-cause), not transmission

This is the conceptual payoff. The within-stratum `Delta > 0` measured by the determinability-boundary work is **observational correlation from a common cause**: within one run, both halves of a stratum share the run's seed and the front's passage, so across an ensemble they co-vary. A1 measures **interventional transmission** and finds it zero. The two diverge *exactly because* the within-stratum correlation is **common-cause, not causal-flow**:

- *Within a stratum:* the two halves are correlated because they **share an origin** (the same run), not because information flows between them.
- *Across the boundary:* independent origins (independent seeds) → no shared cause → no correlation.

So the "severance" the determinability work found is severance of **shared origin** (independent seeds make independent strata), **not** severance of a transmission channel — because there is no transmission channel anywhere. **The determinability boundary is a fact about common causes, not about signals.**

Two notes. *The reframe is descriptive, not a new mechanism:* it re-describes what the boundary *is*, leaving ED's primitives unchanged. *And it is the textbook hallmark of common cause:* observational mutual information can be nonzero while interventional mutual information is exactly zero — correlation without transmission — which is precisely the configuration we measure.

4.3 ED's locality, confirmed as exact channel-zero

A1 demonstrates ED's "no unmediated nonlocal influence" property directly and at the strongest level: **controlled transmission between distinct regions is exactly zero.** Locality here is precise: a node's committed state is a function only of its local neighborhood — the adjacency/transport primitives **P02–P05** — so **no distant intervention can alter it**, which is exactly what the exact channel-zero measures. A committed fact in ED is strictly local. ED's certified locality is not merely an internal code property; it is observable as exact channel-zero, within and across.

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5. Limitations (honest)

- **One certified substrate; specific regions/baselines (5/5) with a passing control.** An exact zero on the cases tested, not a proof for all region geometries.
- **The result is a clean negative** on the capacity rescue (no positive invariant) and a clean positive on the clarification (common-cause, locality). It is not a new positive determinability quantity.
- **The common-cause split is here demonstrated, not formalized.** A transfer-entropy / common-cause decomposition (the interventional term identically zero, all content in the common-cause term) would formalize it; that is a reformulation, not required for the result.

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6. Falsification Criteria

6.1 Nonzero distant-region propagation

Falsifier sentence: *A measurement showing the distant region's final state changes (`L2 > 0`) under a change of the encoded message, within a stratum or across the boundary, would falsify the exact channel-zero result and ED's locality.*

6.2 A positive intrinsic capacity contrast

Falsifier sentence: *A demonstration of a within-stratum channel capacity bounded away from zero (a genuine "within > across" capacity contrast), surviving the local-response and across-null controls, would falsify §4.1 and recover a positive intrinsic determinability invariant.*

6.3 Causal flow within a stratum

Falsifier sentence: *Evidence that the within-stratum `Delta > 0` reflects information flow between the halves (interventional transmission) rather than shared-origin co-variation would falsify the common-cause reframe (§4.2).*

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7. Position Statement

The controlled channel capacity between distinct regions of the certified ED substrate is exactly zero — within strata and across the boundary — by ED's locality.

Capacity therefore yields no positive intrinsic determinability invariant; the only observable-invariant content of the boundary is a zero. The conceptual payoff: the determinability boundary is an **observational (common-cause) structure, not a transmission channel** — the within-stratum correlation is shared-origin co-variation, the across-boundary independence is independent origins, and there is no signal channel anywhere. As a corollary, ED's "no unmediated nonlocal influence" is confirmed as exact channel-zero.

What this paper claims. Exact zero distant-region propagation (5/5, with a passing local control); no positive intrinsic capacity invariant; the boundary's positive content is observational/common-cause; ED's locality is empirically exact.

What this paper does not claim. No positive determinability scalar is found (it is a negative on the rescue); the result is an exact zero on the tested cases, not a geometry-universal proof; the common-cause decomposition is demonstrated, not formalized; locality is confirmed, not derived.

Series context. The robustness companion to the determinability-boundary measurement, and the *internal* reading of the same finite-reach ceiling whose *external* reading is the finite-memory (primes) result: ED's hard ceiling on long-range / global information, measured against ED's own observables here and against a math-certified ruler there.

Appendix — Artifacts and Glossary

Artifacts (`evaluation/Bits/`)

- `analysis/capacity.py` — bias-corrected leave-one-out k-NN decoder MI with shuffle-null (reusable).
- `examples/capacity_experiment.py` — the coding experiment + the exact propagation diagnostic.

Glossary

- **Stratum.** A determinability-equivalence region of the ED substrate — within one run, a connected set of nodes sharing the run's seed and the front's passage, hence co-varying; distinct strata have independent origins.
- **Channel capacity.** The supremum of transmitted information over all inputs and readouts — observable-independent by construction; here, exactly zero between distinct regions.
- **Interventional vs observational.** *Interventional* = change an input and measure transmission (zero here). *Observational* = correlation across an ensemble (the within-stratum `Delta > 0`).
- **Common cause.** Two regions co-vary because they share an origin (a run's seed + front passage), not because a signal flows between them. The determinability boundary is a common-cause structure.
- **Severance.** Independent seeds make independent strata — severance of shared origin, not of a channel (there is none).
- **Locality (ED).** No unmediated nonlocal influence; confirmed here as exact channel-zero — a committed fact depends only on its local neighborhood.

Reader map and open work

Intuition. Two regions in ED can look correlated because they share a run's seed, but no intervention in one can influence the other — exactly the difference between **correlation** and **causation**. The determinability boundary lives entirely on the correlation side.

A note of context. This is the operational content of Pearl's **causal hierarchy**: observational association and interventional effect are distinct rungs, and channel capacity — the interventional measure — is the gold standard for actual causal influence. Here the two rungs come apart maximally: nonzero association (`Delta > 0`), exactly zero intervention.

Where to look next. - *The determinability-boundary measurement this companions:* the Bits determinability paper. - *The external (math-certified) reading of the same ceiling:* the Finite-Memory (primes) paper. - *The coarse-graining wall (the continuum / PDE shadow):* the Continuum paper.

Open work (declared): test more region geometries and partitions; multi-front interaction regimes; whether stochastic ED variants change the picture; and a formal transfer-entropy / common-cause decomposition (interventional term identically zero, all content in the common-cause term).

End of Paper (Common Cause, Not Channel).

Substrate-evaluation arc. Controlled channel capacity between distinct ED regions is exactly zero (`L2 = 0`, 5/5, local control passes) — within strata and across the boundary — by ED's locality. Capacity yields no positive intrinsic determinability invariant; the determinability boundary is an observational (common-cause) structure, not a transmission channel; ED's "no nonlocal influence" is confirmed as exact channel-zero. The boundary's positive content is observational; its interventional content is a universal zero.